LAB 3 — THINKING SHEET

Linear Regression and Stochastic Gradient Descent

This sheet is for you. Preparing for the exam starts here.
I recommend working in pairs, trying to explain your point of view to each other.
For each investigation: PREDICT → RUN & OBSERVE → EXPLAIN → EXPERIMENT
Write short answers. A few words, a sketch, an equation, or a small graph may be enough.
Part 1 (Investigations 1–5): about 50 minutes. Part 2 (Investigations 6–9): about 40 minutes.

1 — Train/test split

PREDICT — before running
Which error do you expect to be smaller, training or test? Must that hold in every experiment?
The training error. No, we could be lucky and have a test set on which the data fit perfectly (even better than the training set)

RUN & OBSERVE → EXPLAIN
Here the file is not sorted by target value, so an unshuffled split would not be a disaster. Why do we shuffle anyway?
Even if the data isn't ordered there might be a pattern that gives us two very different distributions between train and test set

EXPERIMENT
Build the unshuffled 70/30 split.
We observed: _____

2 — Scaling, using training data only

PREDICT — before running
After fitting StandardScaler on the training set, what mean/std do the SCALED TRAINING features have? And the SCALED TEST features?

Training: 0,1 Test: ≈0,1

RUN & OBSERVE → EXPLAIN
The test features do not have mean 0 and std 1. Why is that the correct behaviour and not a bug?
Because train and test set do not share the exact same distribution or size

EXPERIMENT
Fit the scaler on the whole dataset. The code runs. Why is the procedure still wrong, and what does it over-estimate?
Because you're modeling your model over "unknown" data it shouldn't know

3 — The closed-form solution, and what scaling does to it

(?)

PREDICT — before running
$\kappa(X)$ with raw features ≈ $\sim 10^5$ with scaled features ≈ $\sim 10^0$
Will the two coefficient vectors be equal? No
Will the two sets of predictions be equal? Yes

RUN & OBSERVE → EXPLAIN
Different coefficients, identical predictions (to ~10^{-13}). Why are the two models the SAME function?

EXPLAIN
Why can solving the normal equations be less numerically accurate than solving the least-squares problem directly? Relate your answer to the condition numbers of X and X^TX.
Solving the normal equations means computing X^TX and since $K(X^TX) = K(X)^2$ → we're squaring the conditioning factor → our model w^* is much less robust than the one obtained with least squares (where we don't compute X^TX)

COMPARE WITH LAB 2
For the Perceptron, scaling changed the learned model. Why does it not change this one?

4 — RSS, RMSE and R^2

PREDICT — before running
RMSE of the constant predictor $\bar{y}$: _____
R^2 of a model using AT alone: _____
Can R^2 be negative on the test set? _____
Can it be negative on the training set, for our model? _____

RUN & OBSERVE → EXPLAIN
Train RMSE: _____ Test RMSE: _____ Test R^2: _____ Why is the baseline R^2 exactly 0 on training but slightly negative on test?

EXPERIMENT
Drop one feature at a time. Which one costs almost nothing? Why can a feature add little (in terms of predictions) once the others are there?

5 — Feature expansion (optional)

PREDICT — before running
As the degree r grows, how do train R^2, test R^2 and $\kappa(\Psi)$ behave? Which of the three is necessarily monotone, and why?

RUN & OBSERVE → EXPLAIN
With 60 training examples, increasing the polynomial degree may lead to an almost perfect training fit but very poor test predictions. Explain how the number of parameters, the rank of the design matrix, and numerical conditioning contribute to this behavior.

6 - IMPLEMENTING SGD

The update, and your implementation

BEFORE CODING
Write the gradient of $\ell(w,(x,y)) = (\langle w,x\rangle - y)^2$ with respect to w, and then the SGD update:

Compare it with the Perceptron update of Lab 2. What do they have in common? What differs, and why?

PREDICT — before running
After 20 epochs, will SGD return exactly the closed-form solution? Will the recorded training MSE decrease monotonically? Can SGD obtain a lower TRAINING MSE than the exact least-squares solution? Why? Could it have a lower TEST error?

RUN & OBSERVE → EXPLAIN
$\|w_SGD - w^*\|$ = _____ test R^2 of SGD: _____ of the closed form: _____. Why does the loss curve flatten ABOVE the red line instead of reaching it?

EXPLAIN
The test R^2 of the two is essentially identical. What does that say about how accurately we need to minimize the empirical risk?

The step size η

PREDICT — before running
Sketch the loss curve for η far too small, well chosen, and far too large:

RUN & OBSERVE → EXPLAIN
Smallest η that still made real progress in 20 epochs: _____ Largest η that did not blow up: _____

DERIVE
Step A: Write the SGD update for one example. Step B: Let $e_i = \langle w,x_i\rangle - y_i$; substitute the updated weights to find $e_i(\text{new})$. Step C: Find a condition on η for $|e_i(\text{new})| < |e_i|$ when $e_i \neq 0$. Step D: Does this guarantee convergence of the entire SGD run?

Does scaling matter for SGD?

PREDICT — commit before running
Same algorithm, same $\eta = 10^{-3}$, unscaled features. Same model? Slightly worse? Something else?

RUN & OBSERVE → EXPLAIN
Largest η that works on the unscaled features: _____ (predicted by the bound: _____). Scaling left the closed-form model untouched but decides whether SGD runs at all. How can both be true?

CONNECT
Scaling changes both the condition number and the magnitude of feature vectors. Why can both matter for SGD? Is the condition number alone sufficient to choose η?

One experiment of your own (optional)

We changed: _____
We predicted: _____
We observed / explain: _____

END-OF-LAB — CLOSE THE NOTEBOOK

Work individually first; then compare with your partner.
1. Why does scaling leave the closed-form model unchanged but decide whether SGD converges?

2. What does $R^2 = 0$ mean, and why can R^2 be negative on the test set?

3. After polynomial feature expansion, why can we still use linear regression and normal equations, even though the design matrix has changed?